

# Clinician Send-off Data Preparation and Rubric Traceability Report

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## Abstract

This appendix documents the data-preparation pipeline, validation criteria, and artefact provenance for clinician send-off across versions v0.3, v4, and v4.1. The workflow follows a pipeline of baseline freeze, deterministic rubric-based review, and constrained resampling—analogous to multi-stage data construction and systematic validation used in clinically grounded dialogue datasets (e.g. OpenR1-Psy). We specify the operations performed at each stage, the formal criteria and gating strategy used for acceptability, and the reproducibility artefacts that support replication.

**Unit of review.** Throughout this appendix, one *row* means one *case*. For multi-turn studies, the reviewed unit is the case record, not per-turn messages.

## 1 Objective

The preparation programme had three concrete objectives:

1. Freeze a reproducible clinician-ready baseline (`v0.3_postclinician_audit`).
2. Run deterministic cross-study reference review with rubric v2 (v4 review outputs).
3. Produce a fully acceptable Study A variant via deterministic replacement of non-acceptable rows only (v4.1), whilst keeping Study B and Study C unchanged.

## 2 Data Preparation Pipeline

The construction follows a staged pipeline: (1) baseline freeze and manifest contract; (2) line-by-line cross-study review under a fixed rubric; (3) deterministic resampling of non-acceptable Study A rows. Figure 1 summarises inputs, operations, and outputs at each stage.

### 2.1 Stage A: Baseline freeze (v0.3)

Consolidated Study A/B/C payloads and gold artefacts into `benchmark/runtime/data/frozen_splits/v0.3_p`. Snapshot-level `manifest.json` records file-level hashes and row counts. Stage 2 and preflight checks in clinician send-off scripts gate package generation. Primary release anchor: `clinician_readiness_v0.3`.

### 2.2 Stage B: Deterministic reference review (v4)

Implemented line-by-line, rule-based cross-study review over 4,220 rows with word-boundary token matching and rubric v2. Generated intermediate review artefacts in `benchmark/runtime/data/verificatio` from the frozen v0.3 baseline. These v4 outputs are review-stage artefacts; the final released re-mediated snapshot is `v4_1_resampled` under release label `clinician_readiness_v4_2026-02-22`.

**Stage A: Baseline freeze**

Input: Study A/B/C payloads and gold artefacts.

Operations: Consolidate into `frozen_splits/v0.3_postclinician_audit/`; materialise snapshot `manifest.json` (file hashes, row counts); add Stage 2 and preflight gates to send-off scripts.

Output: Frozen snapshot, release package, and package-generation gates.

**Stage B: Cross-study review**

Input: Frozen v0.3 snapshot; rubric v2 (`rubric_rules_v4.json`, word-boundary matching).

Operations: Line-by-line rule-based review over 4,220 rows (Study A: 2,000; Study B single: 2,000; Study B multi: 120; Study C: 100); enforce rubric criteria and token-boundary logic.

Output: SSV verdict files and `run_metadata.json` in `benchmark/runtime/data/verification/v4/`.

**Stage C: Study A resampling (v4.1)**

Input: v4 verdicts; OpenR1-Psy candidate pool; same rubric v2 and scorer.

Operations: Identify non-acceptable Study A rows (NEEDS\_REVIEW + REJECT); replace only those rows using deterministic candidate ordering and in-repo v4 scorer acceptance; preserve Study A IDs and order; leave acceptable rows and Studies B/C untouched.

Output: `frozen_splits/v4_1_resampled`; `benchmark/runtime/data/verification/v4_1` artefacts.

Figure 1: Data preparation pipeline: stages, inputs, operations, and outputs.

### 2.3 Stage C: Deterministic Study A resampling (v4.1)

Identified 768 non-acceptable Study A rows from v4 (761 NEEDS\_REVIEW, 7 REJECT). Replaced only those rows from OpenR1-Psy using deterministic candidate ordering and acceptance with the in-repo v4 scorer. Preserved Study A IDs and order; retained acceptable rows untouched. Produced `frozen_splits/v4_1_resampled` and `verification/v4_1` artefacts.

#### Deterministic resampling procedure.

```

flagged_ids = [study_a ids where verdict in {NEEDS_REVIEW, REJECT}] # source order
used_source_ids = source_openr1_ids from retained ACCEPTABLE rows

for item_id in flagged_ids:
    for candidate in OpenR1Pool(train_then_test, ascending_index):
        if candidate.source_id in used_source_ids:
            continue
        score = score_study_a(candidate, rubric_v2)
        if score.verdict == "ACCEPTABLE":
            replace_row_preserving_id_and_position(item_id, candidate)
            used_source_ids.add(candidate.source_id)
            break

```

**OpenR1 reproducibility pin.** OpenR1-based artefacts in this pipeline are pinned to dataset GMLHUHE/OpenR1-Psy, revision 56fc0ef2fa5926df86713ed9b35f8689a6f85425. Candidate pool for Stage C is iterated in deterministic order (`train` then `test`, ascending dataset index) with eligibility constrained to rows that yield non-empty prompt/answer/reasoning fields via `iter_openr1_candidates()` and `_first_openr1_fields()` in `benchmark/runtime/src/reliable_clinical_`. already-used source IDs are excluded.

## 3 Validation Criteria and Scoring

### 3.1 Verdict space and gate condition

Each reviewed row receives one of three verdicts: `ACCEPTABLE`, `NEEDS_REVIEW`, or `REJECT`. The review logic is deterministic and rule-based. Clinician audit is applied in the baseline freeze process (`v0.3_postclinician_audit`); `v4/v4.1` review and replacement selection are fully automated. Quality is enforced by requiring that published releases use only rows passing the acceptability bar (`v4.1` achieves 100% `ACCEPTABLE` for Study A by replacing non-acceptable rows).

### 3.2 Rubric dimensions (v2)

Canonical rubric version is `v2`. File names `rubric_rules_v4.json` and `rubric_rules_v4_1.json` are historical dataset-version labels that both encode rubric `v2`. These files define criteria for *row acceptability* (not model scoring), summarised below.

Table 1: Rubric v2 operational dimensions (summary).

| Dimension              | Criterion                                                                                                                                                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Matching               | <code>rule_version: v2</code> ; <code>matching_mode: word_boundary</code> ; negation handled within <code>negation_window_chars</code> via <code>negation_phrases</code> .                                                           |
| Completeness           | <code>min_nontrivial_steps</code> , <code>nontrivial_step_min_words</code> ; <code>repetition_threshold</code> to penalise repetitive content.                                                                                       |
| Context alignment      | <code>context_min_prompt_token_matches</code> ; <code>c3_min_overlap_tokens</code> ; <code>context_contradiction_phrases</code> and <code>diagnostic_contradiction_phrases</code> for contradiction detection.                       |
| Framework and symptoms | <code>framework_signals</code> (e.g. DSM, ICD, CBT, ACT, humanistic); <code>symptom_tokens</code> and <code>symptom_token_variants</code> (e.g. eating-disorder variants) with word-boundary matching to avoid false positives.      |
| Risk and safety        | <code>risk_marker_groups</code> (suicidality, self_harm, psychosis, mania, substance, eating_disorder); <code>severe_risk_classes</code> ; <code>safety_ack_tokens</code> ; <code>unsafe_patterns</code> for disqualifying language. |

### 3.3 Gating strategy

The preparation pipeline relies on deterministic, fail-closed checks:

- Snapshot/release manifest integrity (presence and hash verification).
- Unit tests for scoring primitives (token boundaries, risk detection, verdict logic).
- SSV contract tests (strict semicolon-separated values with fixed headers and verdict schema; row counts, uniqueness, valid verdict space, summary consistency).
- Send-off preflight and Stage 2 gates to prevent ambiguous package publication.

## 4 Rubric Evolution and Contract

### 4.1 Canonical rubric inventory

Canonical files: `benchmark/runtime/data/rubrics/rubric_rules_v3.json`, `rubric_rules_v4.json`, `rubric_rules_v4_1.json`.

| File                   | Rule version | Matching mode | SHA256                                        |
|------------------------|--------------|---------------|-----------------------------------------------|
| rubric_rules_v3.json   | v1           | –             | de4af1034cd7fe89ca20c18dd218a86710a03eeb75620 |
| rubric_rules_v4.json   | v2           | word_boundary | e7a453e38bce8aba05997eb619d82534cfc32d648410f |
| rubric_rules_v4_1.json | v2           | word_boundary | e7a453e38bce8aba05997eb619d82534cfc32d648410f |

## 4.2 Operational role of v2

v2 introduced strict word-boundary matching and explicit negation handling, reducing false positives from substring collisions (e.g. *eating* in unrelated contexts). It also expanded framework and risk-marker handling whilst keeping scoring deterministic and local. v4 and v4.1 share the same rubric v2 contract.

## 5 Minimal Rebuild Commands

Code pin for this appendix and command paths: git commit a5e2870 (release label `clinician_readiness_v4_2026-02-22`)

The minimal deterministic replay sequence is:

```
PYTHONPATH=src python scripts/studies/v4_review/run_v4_cross_study_review.py \
--clean --input-root benchmark/runtime/data/frozen_splits/v0.3_postclinician_audit \
--out-dir benchmark/runtime/data/verification/v4 \
--rules benchmark/runtime/data/rubrics/rubric_rules_v4.json

PYTHONPATH=src python scripts/studies/v4_review/run_v4_1_resample_study_a.py --clean

PYTHONPATH=src python scripts/studies/v4_review/run_v4_cross_study_review.py \
--clean --input-root benchmark/runtime/data/frozen_splits/v4_1_resampled \
--out-dir benchmark/runtime/data/verification/v4_1 \
--rules benchmark/runtime/data/rubrics/rubric_rules_v4_1.json

PYTHONPATH=src python -m pytest tests/unit/review/test_v4_reference_review.py \
tests/unit/review/test_v4_ssv_contract.py \
tests/unit/review/test_v4_1_resample.py \
tests/unit/review/test_v4_1_contract.py -q
```

Commands above are pinned to this repository release label (`clinician_readiness_v4_2026-02-22`); paths may differ outside this release line. Determinism check in release workflow: rerun with identical inputs and verify hash-equality for `benchmark/runtime/data/frozen_splits/v4_1_resampled/manifest.json`, `benchmark/runtime/data/verification/v4/study_*_verdicts.ssv`, `benchmark/runtime/data/verification/v4/run_metadata.json`, `benchmark/runtime/data/verification/v4_1/review_summary.ssv`, and `benchmark/runtime/data/verification/v4_1/contract.ssv`.

## 6 Dataset and Run Statistics

Table 3 gives the v4 review summary (run metadata). Table 4 summarises the before/after quality snapshot for Study A.

The v4.1 change was achieved by deterministic replacement of the 768 non-acceptable rows only; retained acceptable rows were not modified. Study B and Study C were unchanged.

## 7 Release-Level Provenance

Table 3: v4 cross-study review summary (from `run_metadata.json`).

| Study            | Rows  | Acceptable | Needs_review | Reject | Duplicate IDs |
|------------------|-------|------------|--------------|--------|---------------|
| Study A          | 2,000 | 1,232      | 761          | 7      | 0             |
| Study B (single) | 2,000 | 2,000      | 0            | 0      | 0             |
| Study B (multi)  | 120   | 120        | 0            | 0      | 0             |
| Study C          | 100   | 100        | 0            | 0      | 0             |
| Total            | 4,220 | 3,452      | 761          | 7      | 0             |

Table 4: Study A quality before and after v4.1 resampling.

|                             | Acceptable | Needs_review | Reject |
|-----------------------------|------------|--------------|--------|
| Pre-resample (v4 review)    | 1,232      | 761          | 7      |
| Post-resample (v4.1 target) | 2,000      | 0            | 0      |

| Release/State            | Primary focus                                  | Key artefacts                                                                                                                    |
|--------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| v0.3_postclinician_ready | Clinician baseline freeze                      | Frozen snapshot payloads and <code>manifest.json</code> ; Stage 2 and preflight gates; clinician package source baseline.        |
| v4_review_outputs        | Deterministic cross-study acceptability review | <code>study_*_verdicts.ssv</code> , <code>review_summary.ssv</code> , <code>run_metadata.json</code> , rubric v2 rule file copy. |
| v4_1_resampled           | Constrained Study A replacement                | Study A deterministic replacement logs, refreshed frozen snapshot, and rerun review outputs at <code>verification/v4_1</code> .  |

## 8 Reproducibility Artefacts

Primary traceability files:

- `benchmark/runtime/data/README.md` (top-level data runbook and canonical rebuild steps)
- `benchmark/runtime/data/rubrics/README.md` (rubric version mapping)
- `benchmark/runtime/data/releases/LATEST.md` (release pointer; current label: `clinician_readiness_v4_1` dataset snapshot state: `v4.1`)
- `benchmark/runtime/data/frozen_splits/*/manifest.json` (snapshot contracts)
- `benchmark/runtime/data/verification/v4*/run_metadata.json` (review run contracts)

## 9 Limitations for Interpretation

- The automated rubric can still be sensitive to lexical edge-cases (for example, idiomatic symptom words), even with word-boundary and negation guards.
- Generation-time output formatting may vary across model backends; post-processing policy should be held constant when comparing runs.
- Inclusion filters in downstream analysis (for example, response-status filters) affect reported aggregates and must be declared with each run.

## 10 Conclusion

The clinician send-off work is documented as an evidence-driven data-engineering pipeline: (1) freeze and validate baseline data contracts; (2) apply deterministic rubric-based review at full cross-study scale; (3) remediate Study A quality via constrained deterministic resampling; and (4) preserve release and artefact-level provenance for reproducibility. The central rubric set is coherent across v3/v4/v4.1, with v4 and v4.1 sharing the same rubric v2 scoring contract.